Oliver Newton

Curriculum Vitæ

Astrophysics Group
Surrey Space Centre, University of Surrey
Guildford, Surrey, GU2 7XH, UK
Musical-Neutron
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Employment & voluntary work

Vocational

2026– **Marie Skłodowska–Curie Fellow**, *Astrophysics Group, University of Surrey*, Guildford, UK.

2022–2026 **Postdoctoral Fellow**, *Center for Theoretical Physics, Polish Academy of Sciences*, Warsaw, Poland.

2022 **PDRA (short-term)**, *Astrophysics Research Institute, Liverpool John Moores University*, Liverpool, UK.

2019–2021 **Postdoctoral researcher**, *Institut de Physique des Deux Infinis*, Lyon, France.

2014–2015 **Graduate Trading Platform Engineer**, *Fidessa*, Woking, UK.

Volunteering

2014–2020 **Founding trustee & Treasurer**, *UniBrass Foundation (charity number:* 1159359*)*, UK.

Education

2015–2019 **PhD**, *Institute for Computational Cosmology*, Durham, UK.
Probing the nature of dark matter with small-scale cosmology
Supervisors: Prof. Adrian Jenkins and Prof. Carlos Frenk

2010–2014 **BSc MPhys**, *University of Warwick*, Coventry, UK, *1st Class (Hons)*.
Masters project: *Determining the fundamental properties of Higgs candidates at the LHC*

Awards and scholarships

2025 **Marie Skłodowska–Curie Postdoctoral Fellowship**, University of Surrey, UK.

2023 **International Office co-financing scheme**, Polish Academy of Sciences, Warsaw, Poland.

2019 **ICC Research Scholarship**, Institute for Computational Cosmology, Durham, UK.

2015–2019 **STFC Postgraduate Studentship**, Institute for Computational Cosmology, Durham, UK.

Competitive computational resource allocations

2026 Awarded 53M CPUh (PI) on DiRAC HPC facility
Awarded 800k CPUh (Co-I) on PLGrid facility

2024 Awarded 2M CPUh (PI) on PLGrid facility

Invited talks and seminars

13 invited talks and seminars.

Oct 2026 **Invited seminar**, University of Liverpool, UK.
High resolution simulations of galaxies as tools to understand the nature of dark matter

May 2025 **Invited seminar**, Waterloo Center for Astrophysics, Canada.
Constraints on the properties of νMSM dark matter using the satellite galaxies of the Milky Way

Sep 2024 **Dark matter & galaxies theory seminar**, Carnegie Theoretical Astrophysics Center, Carnegie Observatories, USA.
Constraints on the properties of νMSM dark matter using the satellite galaxies of the Milky Way

May 2024 **Cosmology and extra-galactic astrophysics seminar**, Department of Physics and Astronomy, UCL, UK.
The formation and disruption of globular clusters in simulations of present-day L^* galaxies with controlled assembly histories

May 2024 **HECA seminar**, Faculty of Physics, University of Warsaw, Warsaw, Poland.
Constraints on the properties of νMSM dark matter using the satellite galaxies of the Milky Way

Feb 2024 **Local Group (online)**, Durham, UK.
Constraining the properties of sterile neutrino DM using the satellite galaxies of the Milky Way

Jan 2024 **CFT colloquium**, Center for Theoretical Physics, Polish Academy of Sciences, Warsaw, Poland.
Exploring the Local Group using constrained simulations

Nov 2022 **Observatory seminar**, Astronomical Observatory, University of Warsaw, Poland.
Exploring the Local Group using constrained simulations

Oct 2022 **Astrophysics group seminar**, Lancaster University, UK.
Exploring the Local Group using constrained simulations

Nov 2021 **Astronomy seminar**, DTU Space, Copenhagen, Denmark.
Hermeian haloes: Field haloes that interacted with the Milky Way and M31

June 2021 **Small-scale structure seminar (online)**, Durham, UK.
Hermeian dark matter haloes of the Local Group

May 2021 **Stars, galaxies and cosmology café club (online)**, LAM Marseilles, France.
Hermeian dark matter haloes of the Local Group

Nov 2020 **Local Group (online)**, Durham, UK.
Constraining the properties of WDM using the satellite galaxies of the Milky Way

Conference contributions

27 contributed talks and four posters.

Contributed talks

Jun 2026 **CLUES Collaboration meeting**, La Laguna, Tenerife, Spain.
The cosmic web of the local Universe

Jun 2025 **European Astronomical Society meeting**, Cork, Ireland.
Constraints on the properties of νMSM dark matter using the satellite galaxies of the Milky Way

Jun 2025 **European Astronomical Society meeting**, Cork, Ireland.
The formation & disruption of globular clusters in simulations of L^* galaxies with controlled assembly histories

Oct 2024 **LSST-Poland meeting**, Warsaw, Poland.
Constraints on the properties of νMSM dark matter using the satellite galaxies of the Milky Way

Jul 2024 **Small Galaxies, Cosmic Questions II**, Durham, UK.
Constraints on the properties of νMSM dark matter using the satellite galaxies of the Milky Way

Jun 2024 **CLUES Collaboration meeting**, Warsaw, Poland.
Constraints on the properties of νMSM dark matter using the satellite galaxies of the Milky Way

Apr 2024 **A new dawn of dwarf galaxy research workshop**, Lorentz Center, Leiden, Netherlands.
The undiscovered ultra-diffuse galaxies of the Local Group

Oct 2023 **LSST-Poland meeting**, Warsaw, Poland.
The undiscovered ultra-diffuse galaxies of the Local Group

Sep 2023 **Polish Astronomical Society Meeting**, Toruń, Poland.
The undiscovered ultra-diffuse galaxies of the Local Group

June 2023 **CLUES Collaboration meeting**, Munich, Germany.
The importance of major mergers for the formation & disruption of globular clusters in present-day L^* galaxies

May 2023 **CACTUS workshop**, London, UK.
Cosmic Web effects on small scales

Sep 2022 **2nd Roman Juszkiewicz Symposium**, Nicolaus Copernicus Astronomical Center, Warsaw, Poland.
The undiscovered ultra-diffuse galaxies of the Local Group

July 2022 **VIRGO Consortium meeting**, MPA Garching, Munich, Germany.
Globular clusters as tracers of galaxy mergers

July 2022 **CLUES Collaboration meeting (online)**, Madrid, Spain.
The undiscovered ultra-diffuse galaxies of the Local Group

Oct 2021 **Świeradów-Zdrój cosmology workshop**, Świeradów-Zdrój, Poland.
Hermeian haloes: Field haloes that interacted with the Milky Way and M31

July 2021 **CLUES Collaboration meeting (online)**, AIP Potsdam, Germany.
Hermeian dark matter haloes of the Local Group

July 2020 **European Astronomical Society meeting (online)**, Leiden, Netherlands.
Constraining the properties of WDM using the satellite galaxies of the Milky Way

Jan 2020 **VIRGO Consortium meeting**, Durham, UK.
Constraining the properties of WDM using the satellite galaxies of the Milky Way

Sep 2019 **CLUES Collaboration meeting**, IN2P3, Lyon, France.
Constraints on thermal relic WDM from satellites of the Local Group

July 2019 **Small Galaxies, Cosmic Questions**, Durham, UK.
Constraints on the mass of the thermal relic warm dark matter particle

Jan 2019 **DEX XV**, Edinburgh, UK.
Constraints on the mass of the thermal relic warm dark matter particle

Dec 2018 **VIRGO Consortium meeting**, Leiden, Netherlands.
Constraints on the mass of the thermal relic warm dark matter particle

Aug 2018 **XXX IAU General Assembly**, Vienna, Austria.
Constraining the mass of the WDM particle using estimates of the total satellite population of the Milky Way

Jan 2018 **DEX XIV**, Durham, UK.
The total satellite population of the Milky Way

Dec 2017 **VIRGO Consortium meeting**, MPA Garching, Munich, Germany.
The total satellite population of the Milky Way

July 2017 **National Astronomy Meeting**, Hull, UK.
The total satellite population of the Milky Way

Dec 2016 **VIRGO Consortium meeting**, Durham, UK.
MW satellite galaxies: how many could there be?

Posters

Jun 2025 **European Astronomical Society meeting**, Cork, Ireland.
The formation & disruption of globular clusters in simulations of L^* galaxies with controlled assembly histories

July 2022 **European Astronomical Society meeting**, Valencia, Spain.
The undiscovered ultra-diffuse galaxies of the Local Group

June 2019 **EWASS**, Lyon, France.
Constraints on the mass of the thermal relic warm dark matter particle

Aug 2018 **XXX IAU General Assembly**, Vienna, Austria.
Constraining the mass of the WDM particle using estimates of the total satellite population of the Milky Way

Supervising experience

PhD supervision

2022–2025 Feven M. Hunde — Center for Theoretical Physics, Polish Academy of Sciences (PAS)
Co-supervised with Dr Wojciech Hellwing and Dr Maciej Bilicki.

Master's supervision

2026– Jakub Możejewski: *The mass profile of the Local Volume from the Timing Argument*
Co-supervised with Dr Mariana Jaber — University of Warsaw, Poland

2026 Zhancheng Ma: *Estimating the satellite galaxy population of the Milky Way* — University of Surrey, UK

2025– Magdaléna Hřibová: *Cosmic web influence on galaxy mergers and properties*
Co-supervised with Dr Mariana Jaber — Masaryk University, Czechia

2017–2019 Co-supervised two 4th year undergraduate students' theses — Durham University, UK

Bachelor's supervision

2024–2025 Jakub Możejewski: *Dark Energy and the Local Universe* — Center for Theoretical Physics, PAS
Co-supervised with Dr Mariana Jaber.

Internships

July–Sep 2024 — Piotr Kuźniar, Michał Czarnowski: *The sensitivity of cosmic web filaments to cosmology*
Co-supervised with Dr. Wojciech Hellwing — Center for Theoretical Physics, PAS

Sep–Nov 2023 — Szymon Bagiński: *The impact of cosmic web environment on dark matter haloes*
Center for Theoretical Physics, PAS

Teaching experience

Dec 2025 — Delivered workshop entitled 'Working with CACTUS' to undergraduate and postgraduate students, postdocs, and faculty — Center for Theoretical Physics, PAS

Sep 2022, Aug 2024 — Delivered a workshop entitled 'Working with simulation outputs in PYTHON' to postgraduate students — Center for Theoretical Physics, PAS

Professional service

Peer review

2024–2025 Natural Sciences and Engineering Research Council (NSERC) of Canada

2022– The Astrophysical Journal (ApJ)

2021– Journal of Cosmology and Astroparticle Physics (JCAP)

Institution

2022–2025 **CCG Group meeting**, *Convener*, CFT-PAN, Poland.

2021 **CLUES discussion meeting**, *Convener*, IP2I Lyon, France.

2016–2017 **Postgraduate Journal Club**, *Co-convener*, Durham, UK.

Conferences and meetings

Jun 2025 **CLUES meeting 2025**, *Session chair*, Potsdam, Germany.

2023–2024 **CLUES meeting 2024**, *Chair of the LOC*, Warsaw, Poland.

Oct 2023 **LSST Poland meeting**, *Session chair*, Warsaw, Poland.

Jun 2023 **CLUES Collaboration meeting**, *Session chair*, Munich, Germany.

2018–2019 **Small Galaxies, Cosmic Questions**, *Member*, Durham, UK.

Memberships

Jun 2022– Junior Member of the International Astronomical Union

Oct 2019– Member CLUES Collaboration

May 2017– Fellow of the Royal Astronomical Society, UK

Oct 2015– Member Virgo Consortium

Outreach

Events & talks

Oct 2026 **World Space Week**, Guildford, UK.
Astrophysics Group

Jul 2026 **Invited talk, Surrey Schools Academy**, Guildford, UK.
Caught in the cosmic web: A pint-sized guide to dark matter

May 2026 **Pint of Science Festival**, Guildford, UK.
Caught in the cosmic web: A pint-sized guide to dark matter

Feb 2026 **Invited talk**, Stonyhurst International School Penang, Malaysia.
Introduction to dark matter, and careers in scientific research

May 2023 **Invited talk**, Caterham School (Online), UK.
Introduction to Dark Matter

Oct 2018 **Celebrate Science**, Durham, UK.
Galaxy Makers

Apr 2018 **Schools Science Festival**, Durham, UK.
Galaxy Makers

Oct 2017 **Celebrate Science**, Durham, UK.
Galaxy Makers

Apr 2017 **Schools Science Festival**, Durham, UK.
Galaxy Makers

Jul 2016 **Royal Society Summer Science Exhibition**, London, UK.
Galaxy Makers

Activity development

2015–2016 **Galaxy Makers**, Durham, UK.
Developed design ideas and ran the constituent EAGLE volumes that were visualised in the final exhibit.

Computing

Languages PYTHON, HPC, LaTeX, TCL, SQL, Microsoft Office suite

Simulations AHF, SubFind, GADGET, AREPO, RAMSES, genetIC

Publications

2025 **Oliver Newton**, Jonathan J Davies, Joel Pfeffer, et al., The formation and disruption of globular cluster populations in simulations of present-day L* galaxies with controlled assembly histories, *MNRAS*, **542**(2):591–607, September 2025, doi: 10.1093/mnras/staf1226

Oliver Newton, Mark R Lovell, Carlos S Frenk, et al., Constraints on the properties of νMSM dark matter using the satellite galaxies of the Milky Way, *MNRAS*, **541**(4):3713–3727, August 2025, doi: 10.1093/mnras/staf1223

Feven Markos Hunde, **Oliver Newton**, Wojciech A. Hellwing, et al., Caught in the cosmic web: Environmental effects on subhalo abundance and internal density profiles, *A&A*, 700:A65, August 2025, doi: 10.1051/0004-6361/202452246

Oliver Newton, The formation and disruption of globular cluster populations in simulations of present-day L* galaxies with controlled assembly histories - Reduced data and scripts, Zenodo, July 2025, doi: 10.5281/zenodo.16614763

2023 Anastasiia Osipova, Sergey Pilipenko, Stefan Gottlöber, et al., Hermeian haloes: Extreme objects with two interactions in the past, *Phys. Dark Universe*, 42:101328, September 2023, doi: 10.1016/j.dark.2023.101328

Oliver Newton, Arianna Di Cintio, Salvador Cardona–Barrero, et al., The Undiscovered Ultradiffuse Galaxies of the Local Group, *ApJL*, **946**(2):L37, March 2023, doi: 10.3847/2041-8213/acc2bb

Oliver Newton, HESTIA Local Group field UDG paper scripts & reduced data, Zenodo, March 2023, doi: 10.5281/zenodo.7757685

2022 **Oliver Newton**, Noam I Libeskind, Alexander Knebe, et al., Hermeian haloes: Field haloes that interacted with both the Milky Way and M31, *MNRAS*, **514**(3):3612–3625, August 2022, doi: 10.1093/mnras/stac1316

Oliver Newton, Hermeian paper plotting code, Zenodo, June 2022, doi: 10.5281/zenodo.6629724

Oliver Newton, Noam I. Libeskind, Alexander Knebe, et al., Hermeian galaxies: Cosmic messengers between the Milky Way and Andromeda, In *High Performance Computing in Science and Engineering*, pages 38–39, Leibniz Rechenzentrumder Bayerische Akademie der Wissenschaften, Garching, 2022, ISBN: 978 3 9816675 5 4

2021 Mark R Lovell, Marius Cautun, Carlos S Frenk, et al., The spatial distribution of Milky Way satellites, gaps in streams, and the nature of dark matter, *MNRAS*, **507**(4):4826–4839, November 2021, doi: 10.1093/mnras/stab2452

Wolfgang Enzi, Riccardo Murgia, **Oliver Newton**, et al., Joint constraints on thermal relic dark matter from strong gravitational lensing, the Ly α forest, and Milky Way satellites, *MNRAS*, **506**(4):5848–5862, October 2021, doi: 10.1093/mnras/stab1960

Oliver Newton, Matteo Leo, Marius Cautun, et al., Constraints on the properties of warm dark matter using the satellite galaxies of the Milky Way, *JCAP*, **2021**(08):062, August 2021, doi: 10.1088/1475-7516/2021/08/062

2018 **Oliver Newton**, Marius Cautun, Adrian Jenkins, et al., The total satellite population of the Milky Way, *MNRAS*, **479**(3):2853–2870, September 2018, doi: 10.1093/mnras/sty1085

Oliver Newton, Marius Cautun, Adrian Jenkins, et al., The Milky Way's total satellite population and constraining the mass of the warm dark matter particle, *Proc. IAU*, **14**(S344):109–113, August 2018, doi: 10.1017/S1743921318006464

Oliver Newton and Marius Cautun, MW Satellite LF: V1.0.0 release, Zenodo, March 2018, doi: 10.5281/zenodo.1205622